

2013

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–17

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which expression is equal to $\int \tan x \, dx$?

(A) $\sec^2 x + c$

(B) $-\ln(\cos x) + c$

(C) $\frac{\tan^2 x}{2} + c$

(D) $\ln(\sec x + \tan x) + c$

2 Which pair of equations gives the directrices of $4x^2 - 25y^2 = 100$?

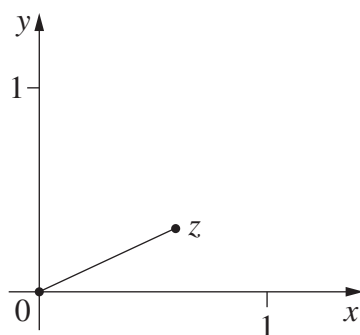
(A) $x = \pm \frac{25}{\sqrt{29}}$

(B) $x = \pm \frac{1}{\sqrt{29}}$

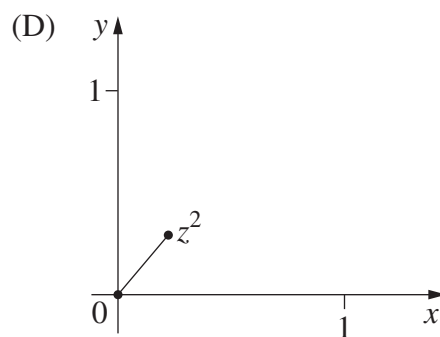
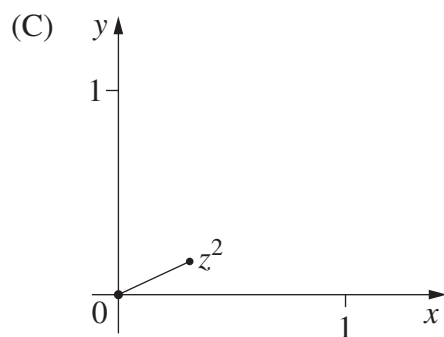
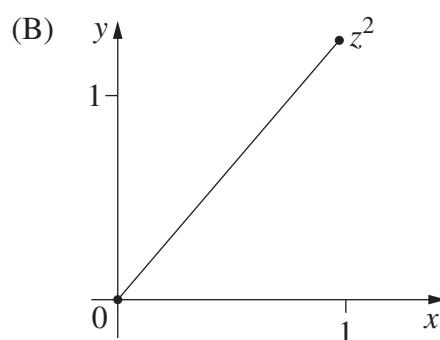
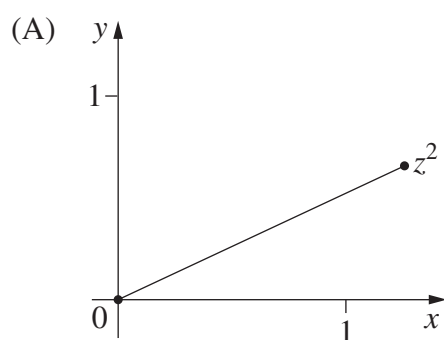
(C) $x = \pm \sqrt{29}$

(D) $x = \pm \frac{\sqrt{29}}{25}$

- 3 The Argand diagram below shows the complex number z .



Which diagram best represents z^2 ?

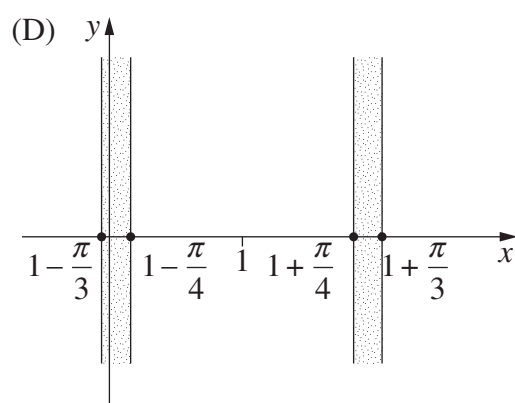
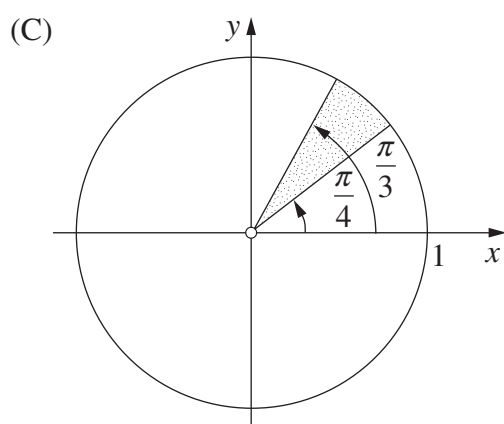
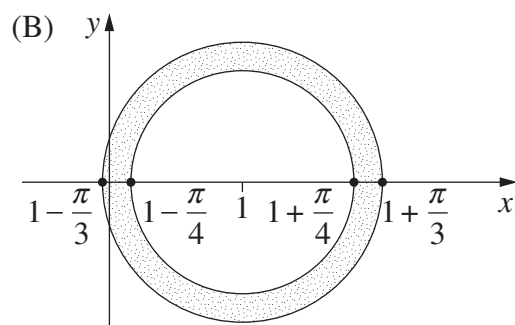
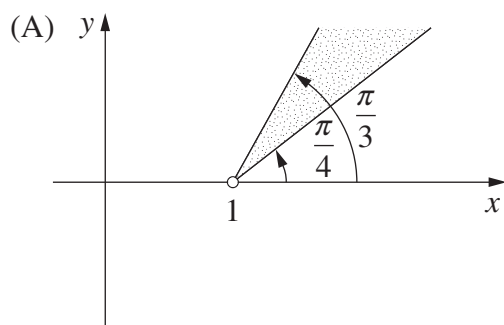


- 4 The polynomial equation $4x^3 + x^2 - 3x + 5 = 0$ has roots α , β and γ .

Which polynomial equation has roots $\alpha + 1$, $\beta + 1$ and $\gamma + 1$?

- (A) $4x^3 - 11x^2 + 7x + 5 = 0$
 (B) $4x^3 + x^2 - 3x + 6 = 0$
 (C) $4x^3 + 13x^2 + 11x + 7 = 0$
 (D) $4x^3 - 2x^2 - 2x + 8 = 0$

- 5 Which region on the Argand diagram is defined by $\frac{\pi}{4} \leq |z-1| \leq \frac{\pi}{3}$?



- 6 Which expression is equal to $\int \frac{1}{\sqrt{x^2 - 6x + 5}} dx$?

(A) $\sin^{-1}\left(\frac{x-3}{2}\right) + C$

(B) $\cos^{-1}\left(\frac{x-3}{2}\right) + C$

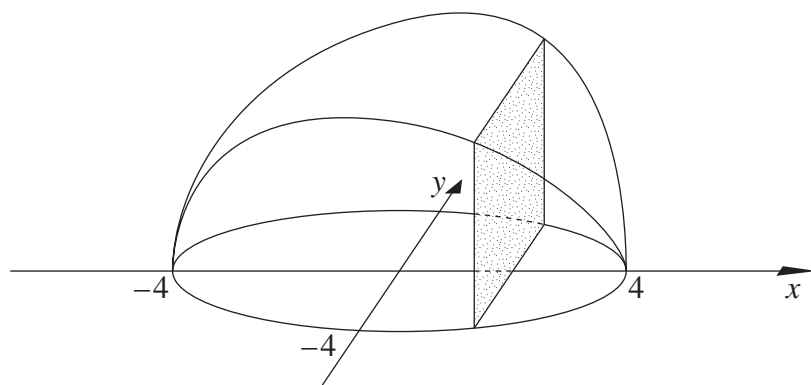
(C) $\ln\left(x-3+\sqrt{(x-3)^2+4}\right) + C$

(D) $\ln\left(x-3+\sqrt{(x-3)^2-4}\right) + C$

- 7 The angular speed of a disc of radius 5 cm is 10 revolutions per minute.

What is the speed of a mark on the circumference of the disc?

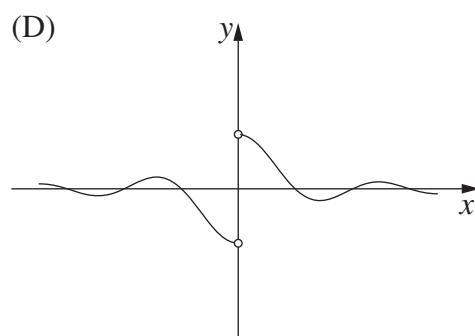
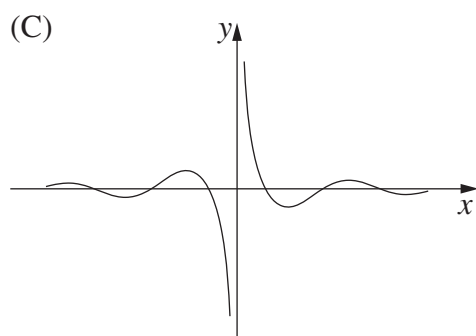
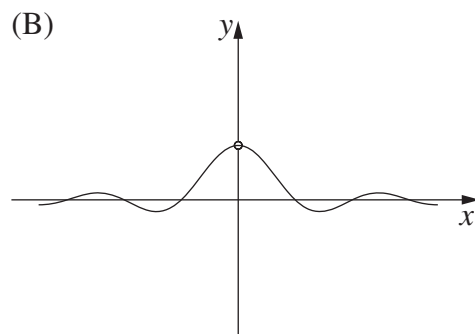
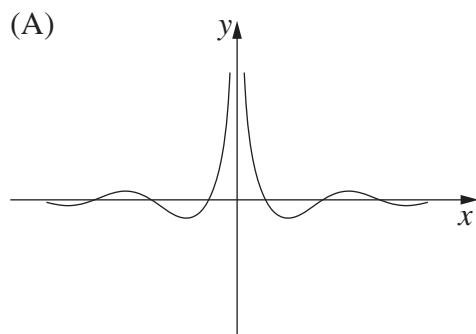
- (A) 50 cm min^{-1}
- (B) $\frac{1}{2} \text{ cm min}^{-1}$
- (C) $100\pi \text{ cm min}^{-1}$
- (D) $\frac{1}{4\pi} \text{ cm min}^{-1}$
- 8 The base of a solid is the region bounded by the circle $x^2 + y^2 = 16$. Vertical cross-sections are squares perpendicular to the x -axis as shown in the diagram.



Which integral represents the volume of the solid?

- (A) $\int_{-4}^4 4x^2 dx$
- (B) $\int_{-4}^4 4\pi x^2 dx$
- (C) $\int_{-4}^4 4(16 - x^2) dx$
- (D) $\int_{-4}^4 4\pi(16 - x^2) dx$

- 9 Which diagram best represents the graph $y = \frac{\sin x}{x}$?



- 10 A hostel has four vacant rooms. Each room can accommodate a maximum of four people.

In how many different ways can six people be accommodated in the four rooms?

- (A) 4020
 (B) 4068
 (C) 4080
 (D) 4096

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = 2 - i\sqrt{3}$ and $w = 1 + i\sqrt{3}$.

(i) Find $z + \bar{w}$. **1**

(ii) Express w in modulus–argument form. **2**

(iii) Write w^{24} in its simplest form. **2**

(b) Find numbers A , B and C such that **2**

$$\frac{x^2 + 8x + 11}{(x - 3)(x^2 + 2)} = \frac{A}{x - 3} + \frac{Bx + C}{x^2 + 2}.$$

(c) Factorise $z^2 + 4iz + 5$. **2**

(d) Evaluate $\int_0^1 x^3 \sqrt{1 - x^2} \, dx$. **3**

(e) Sketch the region on the Argand diagram defined by $z^2 + \bar{z}^2 \leq 8$. **3**

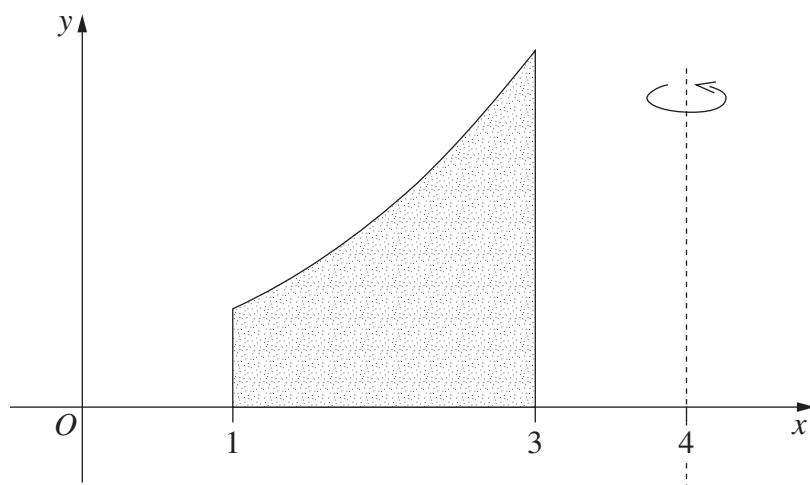
Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{4 + 5 \cos x} dx$. **4**

(b) The equation $\log_e y - \log_e (1000 - y) = \frac{x}{50} - \log_e 3$ implicitly defines y as a function of x . **2**

Show that y satisfies the differential equation $\frac{dy}{dx} = \frac{y}{50} \left(1 - \frac{y}{1000} \right)$.

(c) The diagram shows the region bounded by the graph $y = e^x$, the x -axis and the lines $x = 1$ and $x = 3$. The region is rotated about the line $x = 4$ to form a solid. **4**



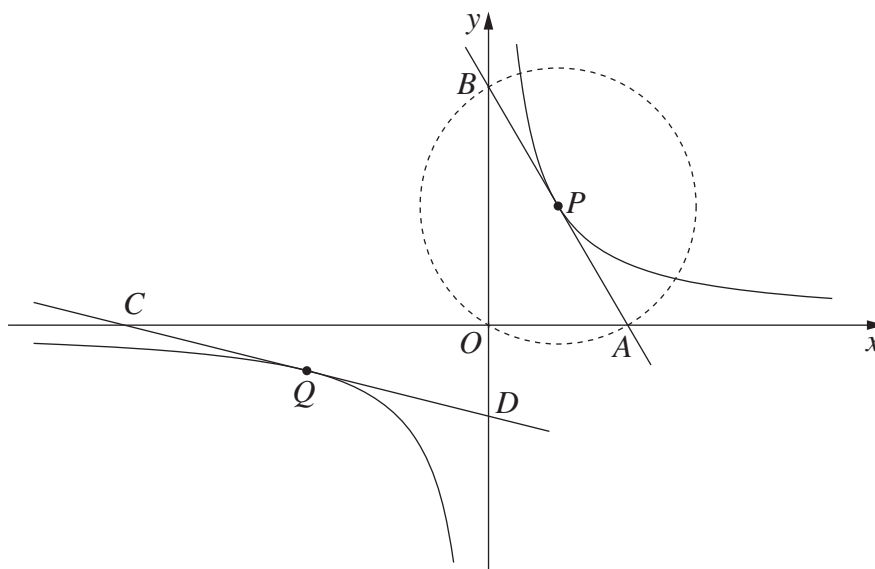
Find the volume of the solid.

Question 12 continues on page 9

Question 12 (continued)

- (d) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where $|p| \neq |q|$, lie on the rectangular hyperbola with equation $xy = c^2$.

The tangent to the hyperbola at P intersects the x -axis at A and the y -axis at B . Similarly, the tangent to the hyperbola at Q intersects the x -axis at C and the y -axis at D .



- | | | |
|-------|--|----------|
| (i) | Show that the equation of the tangent at P is $x + p^2y = 2cp$. | 2 |
| (ii) | Show that A , B and O are on a circle with centre P . | 2 |
| (iii) | Prove that BC is parallel to PQ . | 1 |

End of Question 12

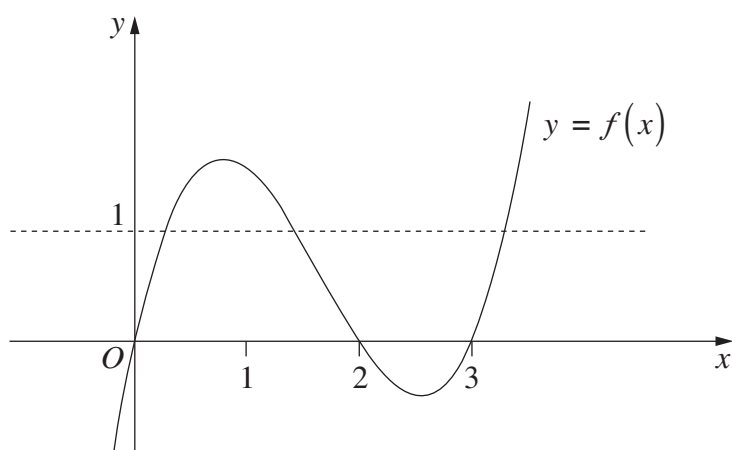
Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Let $I_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$, where $n \geq 0$ is an integer.

(i) Show that $I_n = \frac{n}{n+1} I_{n-2}$ for every integer $n \geq 2$. **3**

(ii) Evaluate I_5 . **2**

(b) The diagram shows the graph of a function $f(x)$.



Sketch the following curves on separate half-page diagrams.

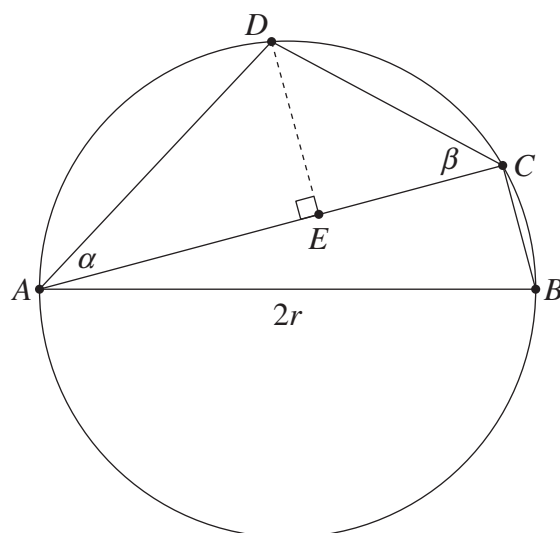
(i) $y^2 = f(x)$ **2**

(ii) $y = \frac{1}{1-f(x)}$ **3**

Question 13 continues on page 11

Question 13 (continued)

- (c) The points A , B , C and D lie on a circle of radius r , forming a cyclic quadrilateral. The side AB is a diameter of the circle. The point E is chosen on the diagonal AC so that $DE \perp AC$. Let $\alpha = \angle DAC$ and $\beta = \angle ACD$.



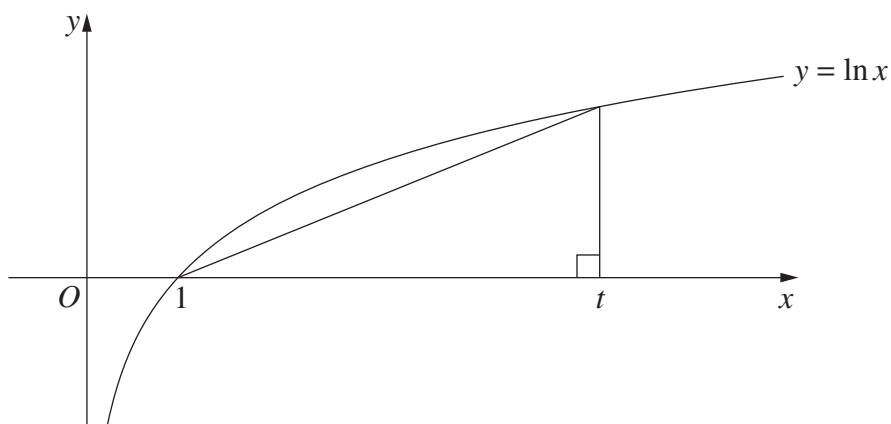
- | | |
|---|----------|
| (i) Show that $AC = 2r \sin(\alpha + \beta)$. | 2 |
| (ii) By considering $\triangle ABD$, or otherwise, show that $AE = 2r \cos \alpha \sin \beta$. | 2 |
| (iii) Hence, show that $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \sin \beta \cos \alpha$. | 1 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the graph $y = \ln x$.

3



By comparing relevant areas in the diagram, or otherwise, show that

$$\ln t > 2 \left(\frac{t-1}{t+1} \right), \text{ for } t > 1.$$

- (b) Let $z_2 = 1 + i$ and, for $n > 2$, let $z_n = z_{n-1} \left(1 + \frac{i}{|z_{n-1}|} \right)$.

3

Use mathematical induction to prove that $|z_n| = \sqrt{n}$ for all integers $n \geq 2$.

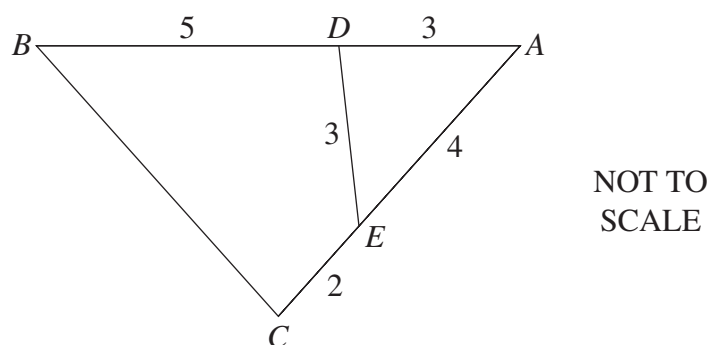
Question 14 continues on page 13

Question 14 (continued)

(c) (i) Given a positive integer n , show that $\sec^{2n} \theta = \sum_{k=0}^n \binom{n}{k} \tan^{2k} \theta$. 1

(ii) Hence, by writing $\sec^8 \theta$ as $\sec^6 \theta \sec^2 \theta$, find $\int \sec^8 \theta d\theta$. 2

- (d) A triangle has vertices A , B and C . The point D lies on the interval AB such that $AD = 3$ and $DB = 5$. The point E lies on the interval AC such that $AE = 4$, $DE = 3$ and $EC = 2$.

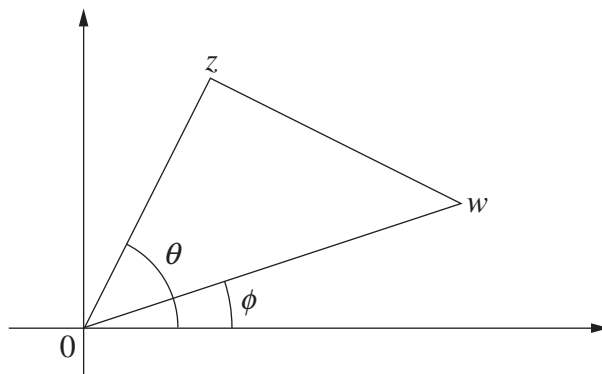


- (i) Prove that $\triangle ABC$ and $\triangle AED$ are similar. 1
- (ii) Prove that $BCED$ is a cyclic quadrilateral. 1
- (iii) Show that $CD = \sqrt{21}$. 2
- (iv) Find the exact value of the radius of the circle passing through the points B , C , E and D . 2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The Argand diagram shows complex numbers w and z with arguments ϕ and θ respectively, where $\phi < \theta$. The area of the triangle formed by 0, w and z is A . **3**



Show that $z\bar{w} - w\bar{z} = 4iA$.

- (b) The polynomial $P(x) = ax^4 + bx^3 + cx^2 + e$ has remainder -3 when divided by $x - 1$. The polynomial has a double root at $x = -1$.
- (i) Show that $4a + 2c = -\frac{9}{2}$. **2**
- (ii) Hence, or otherwise, find the slope of the tangent to the graph $y = P(x)$ when $x = 1$. **1**
- (c) Eight cars participate in a competition that lasts for four days. The probability that a car completes a day is 0.7 . Cars that do not complete a day are eliminated.
- (i) Find the probability that a car completes all four days of the competition. **1**
- (ii) Find an expression for the probability that at least three cars complete all four days of the competition. **2**

Question 15 continues on page 15

Question 15 (continued)

- (d) A ball of mass m is projected vertically into the air from the ground with initial velocity u . After reaching the maximum height H it falls back to the ground. While in the air, the ball experiences a resistive force kv^2 , where v is the velocity of the ball and k is a constant.

The equation of motion when the ball falls can be written as

$$m\dot{v} = mg - kv^2. \quad (\text{Do NOT prove this.})$$

- (i) Show that the terminal velocity v_T of the ball when it falls is $\sqrt{\frac{mg}{k}}$. **1**
- (ii) Show that when the ball goes up, the maximum height H is **3**

$$H = \frac{v_T^2}{2g} \ln \left(1 + \frac{u^2}{v_T^2} \right).$$

- (iii) When the ball falls from height H it hits the ground with velocity w . **2**
- Show that $\frac{1}{w^2} = \frac{1}{u^2} + \frac{1}{v_T^2}$.

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Find the minimum value of $P(x) = 2x^3 - 15x^2 + 24x + 16$, for $x \geq 0$. 2

- (ii) Hence, or otherwise, show that for $x \geq 0$, 1

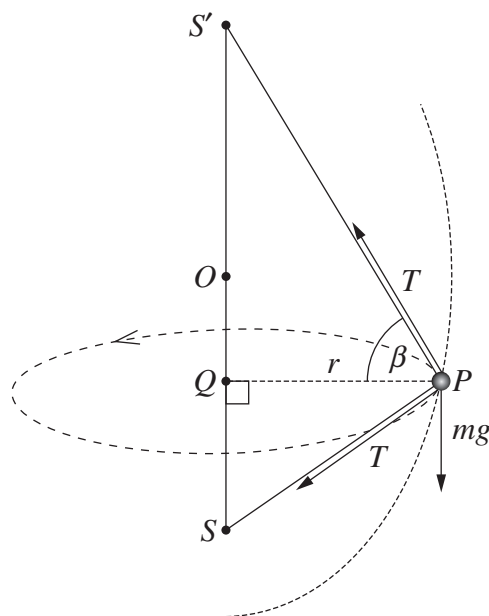
$$(x+1)\left(x^2 + (x+4)^2\right) \geq 25x^2.$$

- (iii) Hence, or otherwise, show that for $m \geq 0$ and $n \geq 0$, 2

$$(m+n)^2 + (m+n+4)^2 \geq \frac{100mn}{m+n+1}.$$

- (b) A small bead P of mass m can freely move along a string. The ends of the string are attached to fixed points S and S' , where S' lies vertically above S . The bead undergoes uniform circular motion with radius r and constant angular velocity ω in a horizontal plane.

The forces acting on the bead are the gravitational force and the tension forces along the string. The tension forces along PS and PS' have the same magnitude T .



The length of the string is $2a$ and $SS' = 2ae$, where $0 < e < 1$. The horizontal plane through P meets SS' at Q . The midpoint of SS' is O and $\beta = \angle S'PQ$. The parameter θ is chosen so that $OQ = a \cos \theta$.

Question 16 continues on page 17

Question 16 (continued)

- (i) What information indicates that P lies on an ellipse with foci S and S' , and with eccentricity e ? **1**
- (ii) Using the focus–directrix definition of an ellipse, or otherwise, show that $SP = a(1 - e \cos \theta)$. **1**
- (iii) Show that $\sin \beta = \frac{e + \cos \theta}{1 + e \cos \theta}$. **2**
- (iv) By considering the forces acting on P in the vertical direction, show that **2**
- $$mg = \frac{2T(1 - e^2)\cos \theta}{1 - e^2 \cos^2 \theta}.$$
- (v) Show that the force acting on P in the horizontal direction is **3**
- $$mr\omega^2 = \frac{2T\sqrt{1 - e^2} \sin \theta}{1 - e^2 \cos^2 \theta}.$$
- (vi) Show that $\tan \theta = \frac{r\omega^2}{g} \sqrt{1 - e^2}$. **1**

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



2013 HSC Mathematics Extension 2 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	A
3	D
4	A
5	B
6	D
7	C
8	C
9	B
10	A

Section II

Question 11 (a) (i)

Criteria	Marks
• Correct answer	1

Sample answer:

$$\begin{aligned} z + \bar{w} &= (2 - i\sqrt{3}) + (1 - i\sqrt{3}) \\ &= 3 - 2\sqrt{3}i \end{aligned}$$

Question 11 (a) (ii)

Criteria	Marks
• Correct solution	2
• Finds modulus or argument	1

Sample answer:

$$|w| = \sqrt{1+3} = 2$$

$$\arg w = \tan^{-1} \frac{\sqrt{3}}{1} = \frac{\pi}{3}$$

Hence $w = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ in modulus–argument form.

Question 11 (a) (iii)

Criteria	Marks
• Correct solution	2
• Attempts to use de Moivre's theorem	1

Sample answer:

By de Moivre's theorem and part (ii)

$$\begin{aligned} w^{24} &= 2^{24} \left(\cos \left(24 \cdot \frac{\pi}{3} \right) + i \sin \left(24 \cdot \frac{\pi}{3} \right) \right) \\ &= 2^{24} (\cos 8\pi + i \sin 8\pi) \\ &= 2^{24} \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Correct solution	2
• Finds value of one of A , B or C	1

Sample answer:

$$\frac{x^2 + 8x + 11}{(x-3)(x^2+2)} = \frac{A}{x-3} + \frac{Bx+C}{x^2+2}$$

$$= \frac{A(x^2+2) + (Bx+C)(x-3)}{(x-3)(x^2+2)}$$

We need: $x^2 + 8x + 11 = A(x^2 + 2) + (Bx + C)(x - 3)$

Make particular choices for x and substitute:

$$x = 3: \quad 44 = 11A$$

$$A = 4$$

$$x = 0: \quad 11 = 2A - 3C$$

$$= 8 - 3C$$

$$C = -1$$

$$x = 1: \quad 20 = 3A - 2B - 2C$$

$$20 = 12 - 2B + 2$$

$$B = -3$$

Hence $A = 4$, $B = -3$ and $C = -1$

Question 11 (c)

Criteria	Marks
• Correct factorisation	2
• Attempts to use formula to solve a quadratic equation, or equivalent merit	1

Sample answer:

Find the zeros by using the quadratic formula:

$$z = \frac{1}{2}(-4i \pm \sqrt{(4i)^2 - 4 \cdot 5 \cdot 1})$$

$$= \frac{1}{2}(-4i \pm \sqrt{-36})$$

$$= -2i \pm 3i$$

The zeros are $-5i$ and i and therefore

$$z^2 + 4iz + 5 = (z + 5i)(z - i)$$

Question 11 (d)

Criteria	Marks
• Correct solution	3
• Correct primitive, or equivalent merit	2
• Uses a suitable substitution to get an indefinite integral, or equivalent merit	1

Sample answer:

Use the substitution

$$x = \sin \theta$$

$$\frac{dx}{d\theta} = \cos \theta$$

When $x = 0$, $\sin \theta = 0$, then $\theta = 0$

When $x = 1$, $\sin \theta = 1$, then $\theta = \frac{\pi}{2}$

Hence

$$\begin{aligned}
 \int_0^1 x^3 \sqrt{1-x^2} \, dx &= \int_0^{\frac{\pi}{2}} \sin^3 \theta \sqrt{1-\sin^2 \theta} \cdot \cos \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{2}} \sin^3 \theta \cos^2 \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{2}} \cos^2 \theta (1 - \cos^2 \theta) \sin \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{2}} \cos^2 \theta \sin \theta - \cos^4 \theta \sin \theta \, d\theta \\
 &= \left[-\frac{\cos^3 \theta}{3} + \frac{\cos^5 \theta}{5} \right]_0^{\frac{\pi}{2}} \\
 &= 0 - \left[-\frac{1}{3} + \frac{1}{5} \right] \\
 &= \frac{2}{15}
 \end{aligned}$$

Alternative: Use the substitution

$$u = 1 - x^2$$

$$\frac{du}{dx} = -2x$$

Where $x = 0$, $u = 1$

When $x = 1$, $u = 0$

$$\begin{aligned}
 \text{Hence } \int_0^1 x^3 \sqrt{1-x^2} \, dx &= -\frac{1}{2} \int_1^0 (1-u) \sqrt{u} \, du \\
 &= \frac{1}{2} \int_0^1 \left(u^{\frac{1}{2}} - u^{\frac{3}{2}} \right) du \\
 &= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_0^1 \\
 &= \frac{2}{15}
 \end{aligned}$$

Question 11 (e)

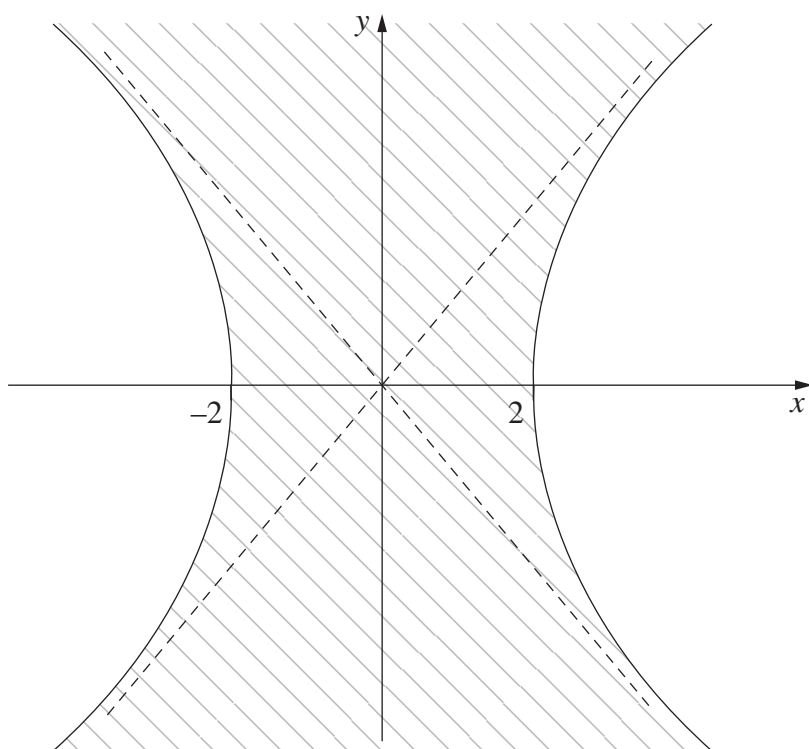
Criteria	Marks
• Correct sketch	3
• Sketches the correct boundary	2
• Obtains $x^2 - y^2 \leq 4$, or equivalent merit	1

Sample answer:

$$z = x + iy \quad (x, y \text{ real})$$

$$\begin{aligned}
 z^2 + \bar{z}^2 &= (x + iy)^2 + (x - iy)^2 \\
 &= x^2 + 2ixy - y^2 + x^2 - 2ixy - y^2 \\
 &= 2x^2 - 2y^2 \\
 &= 2(x^2 - y^2)
 \end{aligned}$$

$$\text{Hence } 2(x^2 - y^2) \leq 8, \text{ so } \frac{x^2}{4} - \frac{y^2}{4} \leq 1.$$



Question 12 (a)

Criteria	Marks
• Correct solution	4
• Makes significant progress towards finding a primitive in terms of t , or equivalent merit	3
• Makes complete substitution including the limits, or equivalent merit	2
• Correctly substitutes for $\cos x$ and attempts to write dx in terms of t , or equivalent merit	1

Sample answer:

First use the t -substitution, $t = \tan \frac{x}{2}$

Then $dx = \frac{2}{1+t^2} dt$ and $\cos x = \frac{1-t^2}{1+t^2}$

When $x = 0$, $t = \tan 0 = 0$ and

when $x = \frac{\pi}{2}$, $t = \tan \frac{\pi}{4} = 1$

Hence $\int_0^{\frac{\pi}{2}} \frac{1}{4+5\cos x} dx$

$$= \int_0^1 \frac{1}{4+5\frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2}{4(1+t^2)+5(1-t^2)} dt$$

$$= \int_0^1 \frac{2}{9-t^2} dt$$

$$= \int_0^1 \frac{2}{(3-t)(3+t)} dt$$

Next, do partial fraction:

$$\frac{2}{(3-t)(3+t)} = \frac{A}{3-t} + \frac{B}{3+t}$$

$$= \frac{A(3+t)+B(3-t)}{(3-t)(3+t)}$$

$$\therefore 2 = A(3+t) + B(3-t)$$

Choose $t = 3$, so $2 = 6A$ and $A = \frac{1}{3}$

Choose $t = -3$, so $2 = 6B$ and $B = \frac{1}{3}$

Hence

$$\int_0^{\frac{\pi}{2}} \frac{1}{4+5\cos x} dx = \frac{1}{3} \int_0^1 \left(\frac{1}{3-t} + \frac{1}{3+t} \right) dt$$

$$= \frac{1}{3} [-\ln(3-t) + \ln(3+t)]_0^1$$

$$= \frac{1}{3} \left[\ln \frac{3+t}{3-t} \right]_0^1$$

$$= \frac{1}{3} \ln 2 - \frac{1}{3} \ln 1$$

$$= \frac{1}{3} \ln 2$$

Question 12 (b)

Criteria	Marks
• Correct solution	2
• Correct implicit differentiation, or equivalent merit	1

Sample answer:

Differentiate both sides of the given identity implicitly with respect to x :

$$\frac{1}{y} \frac{dy}{dx} + \frac{1}{1000 - y} \frac{dy}{dx} = \frac{1}{50}$$

$$\left(\frac{1}{y} + \frac{1}{1000 - y} \right) \frac{dy}{dx} = \frac{1}{50}$$

$$\frac{1000 - \cancel{y} + \cancel{y}}{y(1000 - y)} \frac{dy}{dx} = \frac{1}{50}$$

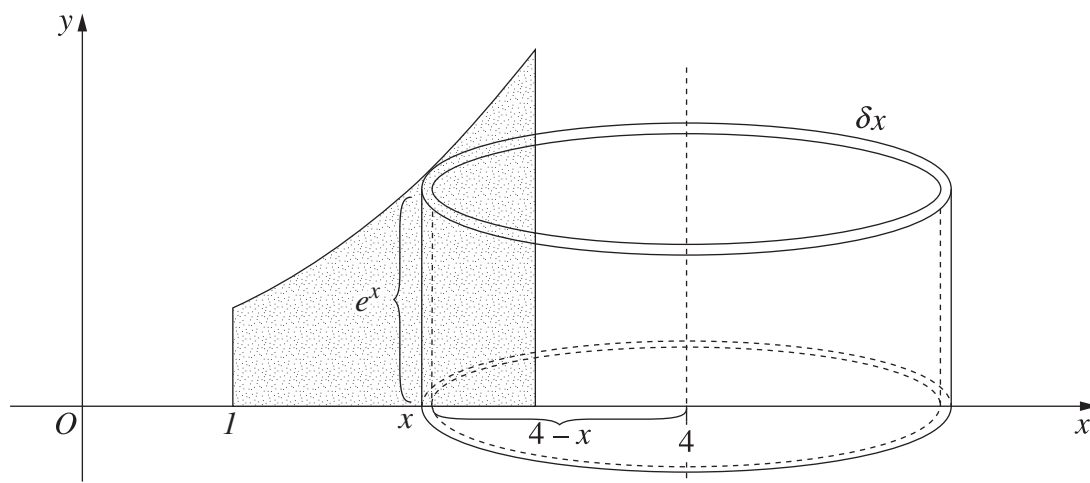
$$\frac{dy}{dx} = \frac{1}{50} y \left(\frac{1000 - y}{1000} \right)$$

$$= \frac{1}{50} y \left(1 - \frac{y}{1000} \right)$$

Question 12 (c)

Criteria	Marks
• Correct solution	4
• Makes substantial progress towards evaluating the integral	3
• Correct integral for volume, or equivalent merit	2
• Attempts to use cylindrical shells, or equivalent merit	1

Sample answer:



Volume of thin cylindrical shell: $2\pi(4-x)e^x\delta x$.

Hence the volume V of the solid formed is

$$\begin{aligned}
 V &= \int_1^3 2\pi(4-x)e^x dx \\
 &= 2\pi \int_1^3 (4e^x - xe^x) dx \\
 &= 2\pi \left[4e^x \right]_1^3 - 2\pi \int_1^3 xe^x dx \\
 &= 8\pi(e^3 - e) - 2\pi \int_1^3 xe^x dx
 \end{aligned}$$

Using integration by parts:

$$\begin{aligned}
 \int_1^3 xe^x dx &= \left[xe^x \right]_1^3 - \int_1^3 e^x dx \\
 &= \left[xe^x - e^x \right]_1^3 \\
 &= (3e^3 - e^3) - (e - e) \\
 &= 3e^3 - e^3 \\
 &= 2e^3
 \end{aligned}$$

$$\text{Hence } V = 8\pi(e^3 - e) - 2\pi(2e^3)$$

$$= 8\pi(e^3 - e) - 4\pi e^3$$

$$= 4\pi e^3 - 8\pi e$$

$$\therefore \text{Volume is } 4\pi e(e^2 - 2) \text{ units}^3$$

Question 12 (d) (i)

Criteria	Marks
• Correct solution	2
• Finds correct gradient, or equivalent merit	1

Sample answer:

$$xy = c^2, \text{ so } y = \frac{c^2}{x}$$

$$\text{Now } \frac{dy}{dx} = -\frac{c^2}{x^2}$$

Hence the slope of the tangent at $\left(cp, \frac{c}{p}\right)$ is

$$-\frac{c^2}{c^2 p^2} = -\frac{1}{p^2}$$

The equation of the tangent is

$$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp$$

Question 12 (d) (ii)

Criteria	Marks
• Correct solution	2
• Finds the coordinates of A and B , or equivalent merit	1

Sample answer:

Find coordinates of A and B using part (i).

$$A : y = 0, \text{ so } x = 2cp \Rightarrow A(cp, 0)$$

$$B : x = 0, \text{ so } p^2 y = 2cp$$

$$y = \frac{2c}{p} \Rightarrow B\left(0, \frac{2c}{p}\right)$$

Now find distances from P :

$$PO : \sqrt{(cp)^2 + \left(\frac{c}{p}\right)^2} = c\sqrt{p^2 + \frac{1}{p^2}}$$

$$PA : \sqrt{(2cp - cp)^2 + \left(\frac{c}{p}\right)^2} = c\sqrt{p^2 + \frac{1}{p^2}}$$

$$PB : \sqrt{(cp)^2 + \left(\frac{2c}{p} - \frac{c}{p}\right)^2} = c\sqrt{p^2 + \frac{1}{p^2}}$$

All distances are equal, so A, B, O are on a circle with centre O .

Question 12 (d) (iii)

Criteria	Marks
• Correct proof	1

Sample answer:

The slope of PQ is

$$m_0 = \frac{\frac{c}{q} - \frac{c}{p}}{cq - cp}$$

$$= \frac{\frac{1}{q} - \frac{1}{p}}{q - p}$$

$$= \frac{p - q}{pq(q - p)}$$

$$= -\frac{1}{pq}$$

Using part (i), the tangent to the hyperbola at Q is $x + q^2y = 2cq$.

At C , $y = 0$ and $x = 2cq$ ie $C(2cq, 0)$.

Hence the slope of BC is

$$m_1 = \frac{\frac{2c}{0} - 0}{0 - 2cq}$$

$$= -\frac{2c}{2cpq}$$

$$= -\frac{1}{pq}$$

As $m_0 = m_1$, the lines PQ and BC are parallel.

Question 13 (a) (i)

Criteria	Marks
• Correct solution	3
• Obtains $I_n = n \int_0^1 x^2 (1-x^2)^{\frac{n-2}{2}} dx$, or equivalent unit	2
• Attempts to use integration by parts, or equivalent merit	1

Sample answer:

Use integration by parts with

$$u = (1-x^2)^{\frac{n}{2}}, \quad v' = 1$$

$$u' = -nx(1-x^2)^{\frac{n}{2}-1}, \quad v = x$$

Hence

$$\begin{aligned}
 I_n &= \left[x(1-x^2)^{\frac{n}{2}} \right]_0^1 + n \int_0^1 x^2 (1-x^2)^{\frac{n}{2}-1} dx \\
 &= 0 - n \int_0^1 \left[(1-x^2) (1-x^2)^{\frac{n}{2}-1} - (1-x^2)^{\frac{n}{2}-1} \right] dx \\
 &= -n \int_0^1 (1-x^2)^{\frac{n}{2}} dx + n \int_0^1 (1-x^2)^{\frac{n-2}{2}} dx \\
 &= -nI_n + nI_{n-2}
 \end{aligned}$$

Solving for I_n

$$(1+n)I_n = nI_{n-2}$$

$$I_n = \frac{n}{n+1} I_{n-2}$$

Question 13 (a) (ii)

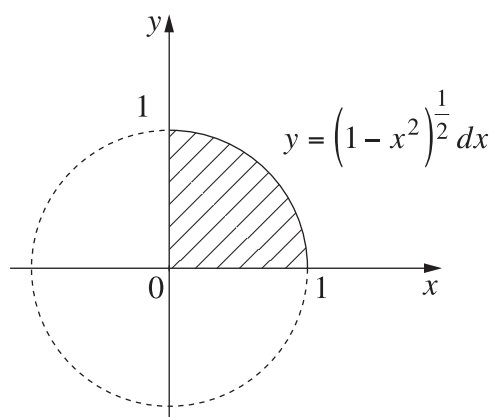
Criteria	Marks
• Correct solution	2
• Finds I_1 , or equivalent merit	1

Sample answer:

$$I_1 = \int_0^1 (1-x^2)^{\frac{1}{2}} dx$$

= area of quarter circle with radius 1

$$= \frac{\pi}{4}$$



From the recursion relation in part (i),

$$I_3 = \frac{3}{4} I_1$$

$$I_5 = \frac{5}{6} I_3$$

$$= \frac{5}{6} \cdot \frac{3}{4} I_1$$

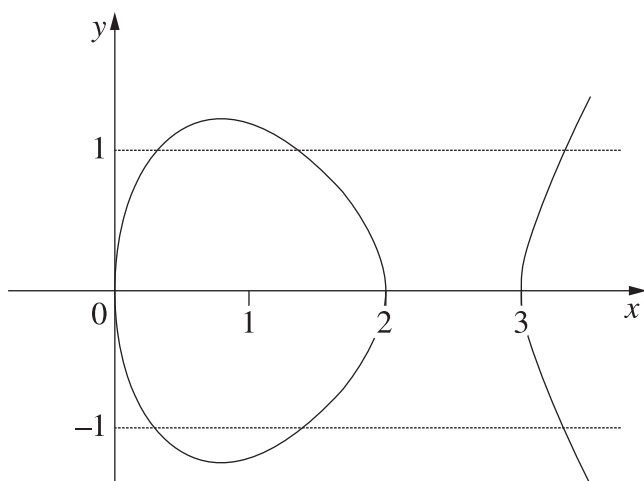
$$= \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{\pi}{4}$$

$$= \frac{5\pi}{32}$$

Question 13 (b) (i)

Criteria	Marks
• Correct sketch	2
• Sketches $y = \sqrt{f(x)}$, or equivalent merit	1

Sample answer:

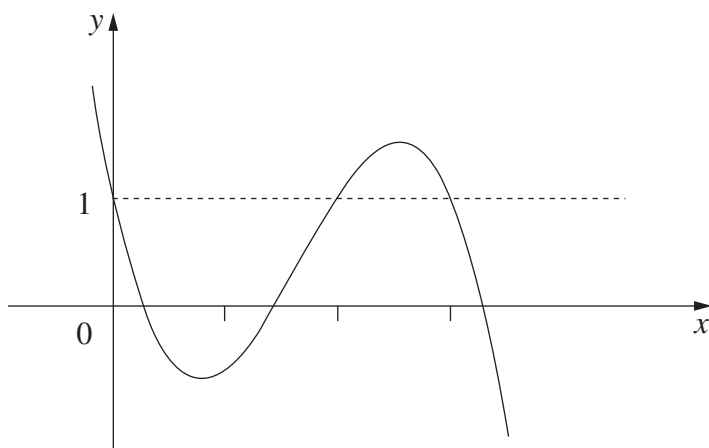


Question 13 (b) (ii)

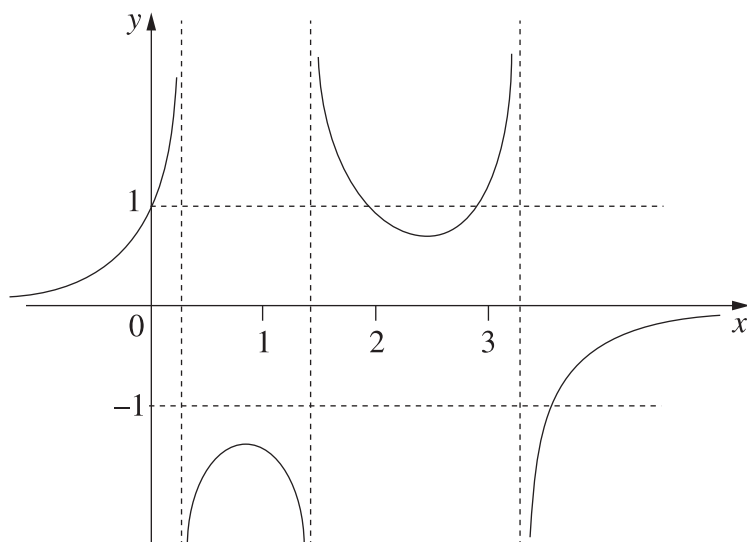
Criteria	Marks
• Correct sketch	3
• Sketches graph showing three correct vertical asymptotes, or equivalent merit	2
• Sketches graph of $y = -f(x)$, or equivalent merit	1

Sample answer:

First sketch $1 - f(x)$



Now sketch $\frac{1}{1 - f(x)}$



Question 13 (c) (i)

Criteria	Marks
• Correct solution	2
• Finds $\angle ABC = \alpha + \beta$, or equivalent merit	1

Sample answer:

$$\angle ADC = 180^\circ - (\alpha + \beta)$$

(angle sum in triangle ACD)

$$\begin{aligned}\angle ABC &= 180^\circ - \angle ADC \\ &= \alpha + \beta\end{aligned}$$

(opposite angles of cyclic quadrilateral)

$$\angle ACB = 90^\circ$$

(angle in semicircle, AB is diameter)

As $\triangle ACB$ is right-angled,

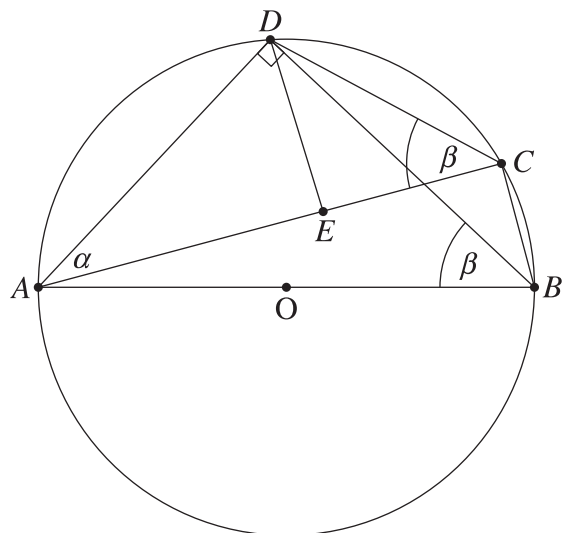
$$\sin(\alpha + \beta) = \frac{AC}{2r}$$

$$\therefore AC = 2r \sin(\alpha + \beta)$$

Question 13 (c) (ii)

Criteria	Marks
• Correct solution	2
• Observes that $\angle ADB$ is a right angle or $\angle ABD = \beta$, or equivalent merit	1

Sample answer:



Join BD .

$$\angle ADB = 90^\circ$$

(angle in semicircle, AB is diameter)

$$\angle ABD = \angle ACD = \beta$$

(angles standing on the same arc AD)

As $\triangle ABD$ is right-angled,

$$\sin \beta = \frac{AD}{AB}$$

$$= \frac{AD}{2r}$$

$$\therefore AD = 2r \sin \beta$$

As $\triangle AED$ is right-angled,

$$\cos \alpha = \frac{AE}{AD}$$

$$\therefore AE = AD \cos \alpha$$

$$= 2r \cos \alpha \sin \beta$$

Question 13 (c) (iii)

Criteria	Marks
• Correct solution	1

Sample answer:

Similarly,

$$CE = 2r \cos \beta \sin \alpha$$

Now

$$AC = AE + CE$$

$$= 2r \cos \alpha \sin \beta + 2r \cos \beta \sin \alpha$$

$$= 2r(\cos \alpha \sin \beta + \cos \beta \sin \alpha)$$

From part (i) $AC = 2r \sin(\alpha + \beta)$, so

$$\sin(\alpha + \beta) = \cos \alpha \sin \beta + \cos \beta \sin \alpha.$$

Question 14 (a)

Criteria	Marks
• Correct solution	3
• Correctly finds both relevant areas, or equivalent merit	2
• Correctly finds one relevant area, or equivalent merit	1

Sample answer:

The area of the triangle with corners $(1, 0)$, $(t, 0)$ and $(t, \ln t)$ is given by $\frac{1}{2}(t-1)\ln t$.

The area under the graph $y = \ln x$ between $x = 1$ and $x = t$ is given by

$$\begin{aligned}
 \int_1^t \ln x \, dx &= \int_1^t 1 \ln x \, dx \\
 &= \left[x \ln x \right]_1^t - \int_1^t x \frac{1}{x} \, dx \quad (\text{integration by parts}) \\
 &= t \ln t - \left[x \right]_1^t \\
 &= t \ln t - (t-1).
 \end{aligned}$$

As the graph $y = \ln x$ is concave down, the area under the graph is greater than the area of the triangle.

$$ie \quad t \ln t - (t-1) > \frac{1}{2}(t-1)\ln t$$

$$t \ln t - \frac{1}{2}(t-1)\ln t > t-1$$

$$\ln t \left[t - \frac{1}{2}(t-1) \right] > t-1$$

$$\ln t \times \frac{1}{2}(t+1) > t-1$$

$$\ln t > 2 \left(\frac{t-1}{t+1} \right), \text{ for } t > 1$$

Question 14 (b)

Criteria	Marks
• Correct solution	3
• Uses the induction hypothesis, or equivalent merit	2
• Establishes initial case	1

Sample answer:

Show true for $n = 2$

$$\begin{aligned}
 |z_2| &= \sqrt{1^2 + 1^2} \\
 &= \sqrt{2} \quad \text{ie true for } n = 2
 \end{aligned}$$

Assume true that $|z_k| = \sqrt{k}$, for some $k \geq 2$

Show true for $n = k + 1$

$$\text{ie } |z_k + 1| = \sqrt{k + 1}$$

$$\begin{aligned}
 \text{LHS} &= |z_{k+1}| = \left| z_k \left(1 + \frac{i}{|z_k|} \right) \right| \\
 &= |z_k| \left| 1 + \frac{i}{|z_k|} \right| \\
 &= \sqrt{k} \left| 1 + \frac{i}{\sqrt{k}} \right| \\
 &= \sqrt{k} \sqrt{1 + \frac{1}{k}} \\
 &= \sqrt{k + 1} \\
 &= \text{RHS}
 \end{aligned}$$

$\therefore$ By the principle of Induction, $|z_n| = \sqrt{n}$ for all integers $n \geq 2$.

Question 14 (c) (i)

Criteria	Marks
• Correct solution	1

Sample answer:

As $\sec^2 \theta = 1 + \tan^2 \theta$ we apply the binomial theorem

$$\begin{aligned}\sec^{2n} \theta &= (\sec^2 \theta)^n \\ &= (1 + \tan^2 \theta)^n \\ &= \sum_{k=0}^n \binom{n}{k} \tan^{2k} \theta\end{aligned}$$

Question 14 (c) (ii)

Criteria	Marks
• Correct solution	2
• Attempts to use part (c) (i), or equivalent merit	1

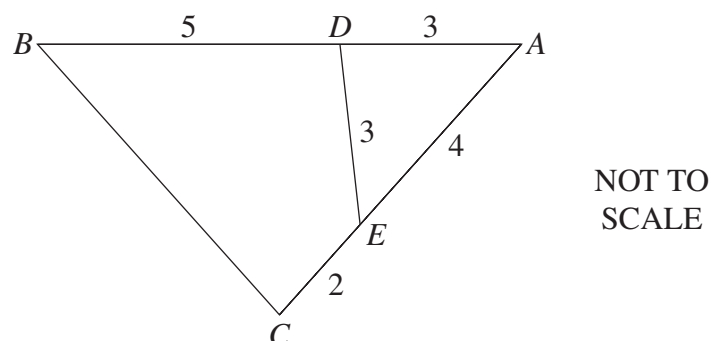
Sample answer:

$$\begin{aligned}\int \sec^8 \theta d\theta &= \int \sec^6 \theta \sec^2 \theta d\theta \\ &= \int \left[\sum_{k=0}^3 \binom{3}{k} \tan^{2k} \theta \right] \sec^2 \theta d\theta \\ &= \sum_{k=0}^3 \binom{3}{k} \int \tan^{2k} \theta \sec^2 \theta d\theta \\ &= \sum_{k=0}^3 \binom{3}{k} \frac{1}{2k+1} \tan^{2k+1} \theta + C \quad \left(\text{since } \frac{d}{d\theta} \tan \theta = \sec^2 \theta \right)\end{aligned}$$

Question 14 (d) (i)

Criteria	Marks
• Correct solution	1

Sample answer:



$\angle DAE$ is common

$$\frac{AB}{AE} = \frac{5+3}{4} = 2 \text{ and}$$

$$\frac{AC}{AD} = \frac{2+4}{3} = 2$$

So $\triangle ABC$ and $\triangle AED$ have pairs of sides in the same ratio, and the included angles equal.
Hence $\triangle ABC$ and $\triangle AED$ are similar.

Question 14 (d) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

$\angle ADE = \angle BCE$ (corresponding angles in similar triangles $\triangle ABC$ and $\triangle AED$)

$$\angle BDE = 180^\circ - \angle ADE$$

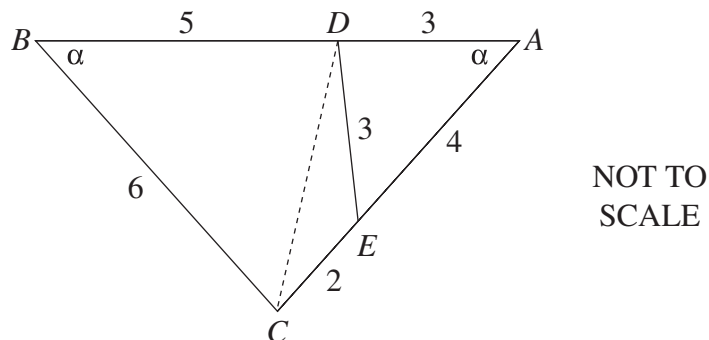
$$= 180^\circ - \angle BCE$$

Hence $BCED$ is cyclic (opposite angles are supplementary).

Question 14 (d) (iii)

Criteria	Marks
• Correct solution	2
• Uses cosine rule in a suitable triangle, or equivalent merit	1

Sample answer:



$\triangle AED$ is isosceles, so by similarity with $\triangle ABC$, $\triangle ABC$ is also isosceles with
 $BC = AC = 4 + 2 = 6$ and
 $\angle ABC = \angle BAC = \alpha$

Applying the cosine rule in $\triangle ADC$

$$CD^2 = 3^2 + 6^2 - 2 \times 3 \times 6 \cos \alpha$$

Applying the cosine rule in $\triangle BDC$

$$CD^2 = 5^2 + 6^2 - 2 \times 5 \times 6 \cos \alpha$$

Eliminate $\cos \alpha$ from the two equations:

$$CD^2 = 45 - 36 \cos \alpha \Rightarrow \cos \alpha = \frac{45 - CD^2}{36}$$

$$\begin{aligned} CD^2 &= 61 - 60 \cos \alpha \\ &= 61 - 60 \left(\frac{45 - CD^2}{36} \right) \\ &= 61 - 75 + \frac{5}{3} CD^2 \end{aligned}$$

$$\frac{2}{3} CD^2 = 14$$

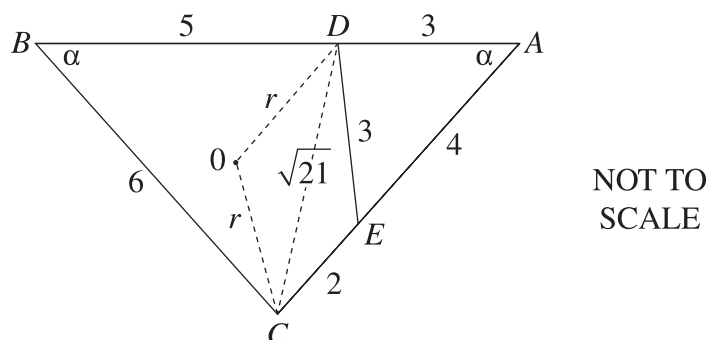
$$CD^2 = 21$$

$$CD = \sqrt{21}$$

Question 14 (d) (iv)

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Sample answer:



Let O be the centre of the circle passing through B , C , E and D ($BCED$ is a cyclic quadrilateral).

$\angle ABC = \alpha$, then $\angle COD = 2\alpha$ (angle at the centre is twice the angle on the circumference standing on the same arc).

$$\text{From part (iii) } \cos \alpha = \frac{45 - CD^2}{36} = \frac{45 - 21}{36} = \frac{2}{3}$$

$$\text{Now } \cos 2\alpha = 2\cos^2 \alpha - 1$$

$$= 2\left(\frac{2}{3}\right)^2 - 1$$

$$= -\frac{1}{9}$$

From the cosine rule in $\triangle CDO$

$$CD^2 = r^2 + r^2 - 2r^2 \cos 2\alpha$$

$$\therefore 21 = 2r^2 + \frac{2}{9}r^2$$

$$= \frac{20}{9}r^2$$

$$\text{Hence } r^2 = \frac{9 \times 21}{20} \text{ and } r = \frac{3\sqrt{21}}{2\sqrt{5}}$$

Question 15 (a)

Criteria	Marks
• Correct solution	3
• Shows that $z\bar{w} - w\bar{z} = 2i w z (\sin\theta\cos\phi - \cos\theta\sin\phi)$, or equivalent merit	2
• Shows the fact that $z = z (\cos\theta + i\sin\theta)$, or equivalent merit	1

Sample answer:

$$\text{Area } \triangle = \frac{1}{2}ab\sin C$$

$$\text{ie } A = \frac{1}{2}|z||w|\sin(\theta - \phi)$$

$$2A = |z||w|\sin(\theta - \phi)$$

$$\begin{aligned} z\bar{w} - w\bar{z} &= |z|(\cos\theta + i\sin\theta)|w|(\cos\phi - i\sin\phi) - |w|(\cos\phi + i\sin\phi)|z|(\cos\theta - i\sin\theta) \\ &= |w||z|(2i\cos\phi\sin\theta - 2i\cos\theta\sin\phi) \\ &= 2i|w||z|(\sin\theta\cos\phi - \cos\theta\sin\phi) \\ &= 2i|w||z|(\sin(\theta - \phi)) \\ &= 2i2A \\ &= 4iA \end{aligned}$$

Question 15 (b) (i)

Criteria	Marks
• Correct solution	2
• Correctly finds the value of b , or equivalent merit	1

Sample answer:

$$P(1) = a + b + c + e = -3 \text{ (remainder theorem)}$$

$$P(-1) = a - b + c + e = 0 \text{ (as } x = -1 \text{ is a root)}$$

Subtracting

$$2b = -3, \text{ so } b = -\frac{3}{2}$$

$$P'(x) = 4ax^3 + 3bx^2 + 2cx$$

$$P'(-1) = -4a + 3b - 2c$$

$$P'(-1) = 0 \text{ (as } x = -1 \text{ is a double root)}$$

Hence

$$4a + 2c = 3b$$

$$= 3\left(-\frac{3}{2}\right)$$

$$= -\frac{9}{2}.$$

Question 15 (b) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

Slope of tangent at $x = 1$ is

$$P'(1) = 4a + 3b + 2c$$

$$= (4a + 2c) + 3b$$

$$= -\frac{9}{2} + 3\left(-\frac{3}{2}\right)$$

$$= -9.$$

Question 15 (c) (i)

Criteria	Marks
• Correct solution	1

Sample answer:

The probability that a car completes all 4 days is

$$(0.7)^4 \approx 0.2401$$

$$= 0.24$$

Question 15 (c) (ii)

Criteria	Marks
• Correct solution	2
• Demonstrates some understanding of binomial probability using $n = 8$, or equivalent merit	1

Sample answer:

The probability that a car does not complete all 4 days is

$$1 - (0.7)^4 \approx 0.76$$

Probability that no car completes all 4 days:

$$\binom{8}{0}(0.76)^8$$

Probability that precisely one completes all 4 days:

$$\binom{8}{1}(0.76)^7(0.24)^1$$

Probability that precisely two complete all 4 days:

$$\binom{8}{2}(0.76)^6(0.24)^2$$

Therefore the probability that 0, 1 or 2 complete all 4 days:

$$(0.76)^8 \left((0.76)^2 + 8(0.76)(0.24) + 28(0.24)^2 \right) \approx 0.7$$

Hence the probability that at least 3 complete all 4 days is

$$1 - 0.7 = 0.3$$

Question 15 (d) (i)

Criteria	Marks
• Correct solution	1

Sample answer:

Terminal velocity occurs when acceleration is zero ($\dot{v} = 0$),

$$\text{ie } mg = kv^2$$

$$v_T^2 = \frac{mg}{k}$$

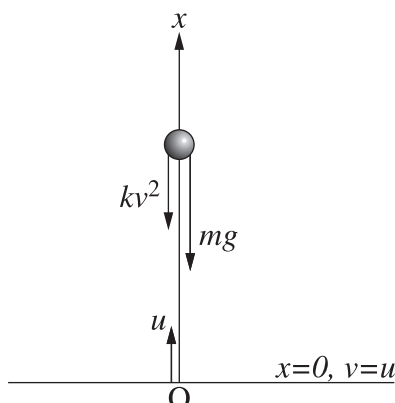
$$v_T = \pm \sqrt{\frac{mg}{k}}$$

$$\text{ie } v_T = \sqrt{\frac{mg}{k}}$$

Question 15 (d) (ii)

Criteria	Marks
• Correct solution	3
• Finds an expression for x in terms of v and an undetermined constant, or equivalent merit	2
• Obtains equation of motion, or equivalent merit	1

Sample answer:



Equation of motion for the ball going up:

$$m\dot{v} = -mg - kv^2$$

$$= -mg \left(1 + \frac{k}{mg} v^2 \right)$$

$$= -mg \left(1 + \frac{v^2}{v_T^2} \right)$$

$$\therefore \dot{v} = \frac{-g}{v_T^2} (v_T^2 + v^2)$$

We know $\dot{v} = \frac{1}{2} \frac{dv^2}{dx}$, so

$$\frac{1}{2} \frac{dv^2}{dx} = -\frac{g}{v_T^2} (v_T^2 + v^2)$$

$$\text{or } \frac{dx}{dv^2} = -\frac{v_T^2}{2g} \left(\frac{1}{v_T^2 + v^2} \right)$$

Integrating we get

$$x = -\frac{v_T^2}{2g} \ln(v_T^2 + v^2) + C$$

When $t = 0$, $x = 0$ and $v = u$. Hence

$$0 = -\frac{v_T^2}{2g} \ln(v_T^2 + u^2) + C$$

$$\text{so } C = \frac{v_T^2}{2g} \ln(v_T^2 + u^2)$$

$$\therefore x = \frac{v_T^2}{2g} \left[\ln(v_T^2 + u^2) - \ln(v_T^2 + v^2) \right]$$

$$= \frac{v_T^2}{2g} \ln \left(\frac{v_T^2 + u^2}{v_T^2 + v^2} \right)$$

When the ball reaches its maximum height,

$$v = 0 \quad \text{and} \quad x = H. \quad \text{Hence}$$

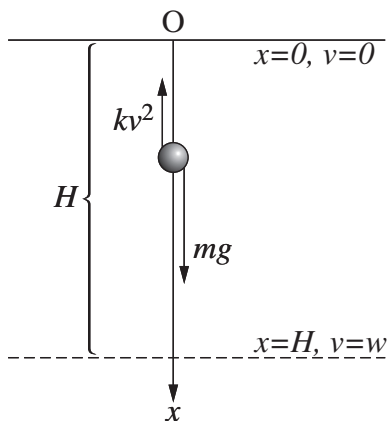
$$H = \frac{v_T^2}{2g} \ln \frac{v_T^2 + u^2}{v_T^2}$$

$$= \frac{v_T^2}{2g} \ln \left(1 + \frac{u^2}{v_T^2} \right)$$

Question 15 (d) (iii)

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Sample answer:



Equation of motion for the ball coming down:

$$\begin{aligned}
 m\dot{v} &= mg - kv^2 \\
 &= mg \left[1 - \frac{k}{mg} v^2 \right] \\
 &= mg \left(1 - \frac{v^2}{v_T^2} \right)
 \end{aligned}$$

$$\therefore \dot{v} = \frac{g}{v_T^2} (v_T^2 - v^2)$$

$$\text{We know } \dot{v} = \frac{1}{2} \frac{dv^2}{dx},$$

$$\text{so } 2 \frac{dx}{dv^2} = \frac{v_T^2}{g} \left(\frac{1}{v_T^2 - v^2} \right)$$

$$\text{Hence } x = \frac{-v_T^2}{2g} \ln(v_T^2 - v^2) + C$$

$$\text{When } x = 0, v = 0,$$

$$\therefore C = \frac{v_T^2}{2g} \ln v_T^2$$

$$x = \frac{v_T^2}{2g} \left[\ln v_T^2 - \ln(v_T^2 - v^2) \right]$$

$$\text{ie } x = \frac{v_T^2}{2g} \ln \left(\frac{v_T^2}{v_T^2 - v^2} \right)$$

When the ball hits the ground, $x = H, v = w$

$$H = \frac{v_T^2}{2g} \ln \left(\frac{v_T^2}{v_T^2 - w^2} \right)$$

$$\text{but } H = \frac{v_T^2}{2g} \ln \left(1 + \frac{u^2}{v_T^2} \right) \quad (\text{from part (ii)})$$

$$\therefore \frac{v_T^2}{2g} \ln \left(1 + \frac{u^2}{v_T^2} \right) = \frac{v_T^2}{2g} \ln \left(\frac{v_T^2}{v_T^2 - w^2} \right)$$

$$1 + \frac{u^2}{v_T^2} = \frac{v_T^2}{v_T^2 - w^2}$$

$$\frac{v_T^2 + u^2}{v_T^2} = \frac{v_T^2}{v_T^2 - w^2}$$

$$(v_T^2 + u^2)(v_T^2 - w^2) = v_T^2 \cdot v_T^2$$

$$\cancel{v_T^2} \cdot \cancel{v_T^2} - v_T^2 w^2 + u^2 v_T^2 - u^2 w^2 = \cancel{v_T^2} \cdot \cancel{v_T^2}$$

$$(\text{dividing by } u^2 w^2 v_T^2) \quad \frac{-1}{u^2} + \frac{1}{w^2} + \frac{1}{v_T^2} = 0$$

$$\text{ie } \frac{1}{w^2} = \frac{1}{u^2} + \frac{1}{v_T^2}$$

Question 16 (a) (i)

Criteria	Marks
• Correct solution	2
• Solves $P'(x) = 0$, or equivalent merit	1

Sample answer:

$$P(x) = 2x^3 - 15x^2 + 24x + 16$$

$$P'(x) = 6x^2 - 30x + 24 = 0 \quad (\text{for max/min})$$

$$\text{ie } x^2 - 5x + 4 = 0$$

$$(x - 4)(x - 1) = 0$$

$$x = 4 \text{ or } 1$$

$$P''(x) = 12x - 30$$

$$P''(4) = 48 - 30 > 0 \therefore \text{min}$$

$$P''(1) = 12 - 30 < 0 \therefore \text{max}$$

$$P(4) = 0 \therefore \text{min at } (4, 0)$$

$$\text{Test boundary: } x = 0, \quad P(0) = 16$$

$\therefore$ minimum value of $P(x)$ is 0 when $x = 4$

Question 16 (a) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

$$(x + 1)(x^2 + (x + 4)^2) \geq 25x^2$$

$$\text{LHS} = (x + 1)(x^2 + x^2 + 8x + 16)$$

$$= 2x^2 + 8x^2 + 16x + 2x^2 + 8x + 16$$

$$= 2x^3 + 10x^2 + 24x + 16$$

Since $P(x) \geq 0$, ie $2x^3 - 15x^2 + 24x + 16 \geq 0$ (from part (i))

$$\text{then } 2x^3 + 10x^2 + 24x + 16 \geq 25x^2$$

$$\text{ie } (x + 1)(x^2 + (x + 4)^2) \geq 25x^2.$$

Question 16 (a) (iii)

Criteria	Marks
• Correct solution	2
• Shows that $(m+n)^2 + (m+n+4)^2 \geq \frac{25(m+n)^2}{m+n+1}$, or equivalent merit	1

Sample answer:

$$(m+n)^2 + (m+n+4)^2 \geq \frac{100mn}{m+n+1}$$

From part (ii)

$$(x+1)(x^2 + (x+4)^2) \geq 25x^2$$

Since $x \geq 0$ then

$$x^2 + (x+4)^2 \geq \frac{25x^2}{x+1}$$

Let $x = m+n$

$$(m+n)^2 + (m+n+4)^2 \geq \frac{25(m+n)^2}{m+n+1}$$

$$\text{Now } (m-n)^2 \geq 0$$

$$m^2 - 2mn + n^2 \geq 0$$

$$m^2 + n^2 \geq 2mn$$

$$\text{and } m^2 + 2mn + n^2 \geq 2mn + 2mn$$

$$(m+n)^2 \geq 4mn$$

$$\begin{aligned} \therefore (m+n)^2 + (m+n+4)^2 &\geq \frac{25(4mn)}{m+n+1} \\ &\geq \frac{100mn}{m+n+1}. \end{aligned}$$

Question 16 (b) (i)

Criteria	Marks
• Correct answer	1

Sample answer:

The string has length $2a$, so $PS' + PS = 2a$.

This characterises an ellipse with focal length a and focal points S and S' .

Also, for an ellipse, $SS' = 2ae$, which is given.

Question 16 (b) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

For an ellipse, the distance from the centre O to the directrix is $\frac{a}{e}$.

Hence the distance from P to the directrix is

$$\frac{a}{e} - OQ = \frac{a}{e} - a \cos \theta.$$

By the focus-directrix definition of an ellipse, $SP = ePM$ (where M is on the directrix)

$$\therefore SP = e \left(\frac{a}{e} - a \cos \theta \right)$$

$$\text{So, } SP = a - ae \cos \theta$$

Question 16 (b) (iii)

Criteria	Marks
• Correct solution	2
• Finds the length of $S'P$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \sin \beta &= \frac{QS'}{PS'} \\
 &= \frac{ae + OQ}{2a - SP} \\
 &= \frac{ae + a \cos \theta}{2a - a(1 - e \cos \theta)} \\
 &= \frac{a(e + \cos \theta)}{2a - a + ae \cos \theta} \\
 &= \frac{a(e + \cos \theta)}{a(1 + e \cos \theta)} \\
 &= \frac{e + \cos \theta}{1 + e \cos \theta}.
 \end{aligned}$$

Question 16 (b) (iv)

Criteria	Marks
• Correct solution	2
• Finds the component of one tension force in the vertical direction, or equivalent merit	1

Sample answer:

The vertical component of the force along $S'P$ is

$$T \sin \beta = T \frac{e + \cos \theta}{1 + e \cos \theta}$$

Now

$$\begin{aligned} \sin(\angle QPS) &= \frac{QS}{PS} \\ &= \frac{ae - OQ}{a(1 - e \cos \theta)} \\ &= \frac{ae - a \cos \theta}{a(1 - e \cos \theta)} \\ &= \frac{e - \cos \theta}{1 - e \cos \theta} \end{aligned}$$

The vertical component of the force along SP is therefore

$$T \sin(\angle QPS) = T \frac{e - \cos \theta}{1 - e \cos \theta}$$

Therefore

$$\begin{aligned} mg &= T \left(\frac{e + \cos \theta}{1 + e \cos \theta} - \frac{e - \cos \theta}{1 - e \cos \theta} \right) \\ &= T \frac{(e + \cos \theta)(1 - e \cos \theta) - (e - \cos \theta)(1 + e \cos \theta)}{1 - e^2 \cos^2 \theta} \\ &= \frac{T}{1 - e^2 \cos^2 \theta} [e + \cos \theta - e^2 \cos \theta - e \cos^2 \theta - (e - \cos \theta + e^2 \cos \theta - e \cos^2 \theta)] \\ &= \frac{T}{1 - e^2 \cos^2 \theta} [2 \cos \theta - 2e^2 \cos \theta] \\ &= \frac{2T(1 - e^2) \cos \theta}{1 - e^2 \cos^2 \theta}. \end{aligned}$$

Question 16 (b) (v)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Finds QP , or equivalent merit	1

Sample answer:

As P lies on an ellipse of eccentricity e

$$r = b = QP = a\sqrt{1-e^2} \sin \theta$$

Since ellipse has $b^2 = a^2(1-e^2)$

$$\begin{aligned} \cos(\angle QPS) &= \frac{r}{SP} \\ &= \frac{\cancel{a}\sqrt{1-e^2} \sin \theta}{\cancel{a}(1-e \cos \theta)} \\ &= \frac{\sqrt{1-e^2} \sin \theta}{1-e \cos \theta} \end{aligned}$$

$$\begin{aligned} \cos(\beta) &= \frac{r}{S'P} \\ &= \frac{\cancel{a}\sqrt{1-e^2} \sin \theta}{\cancel{a}(1+e \cos \theta)} \\ &= \frac{\sqrt{1-e^2} \sin \theta}{1+e \cos \theta} \end{aligned}$$

Hence the horizontal force is

$$\begin{aligned} mrw^2 &= \frac{T\sqrt{1-e^2} \sin \theta}{1-e \cos \theta} + \frac{T\sqrt{1-e^2} \sin \theta}{1+e \cos \theta} \\ &= \frac{2T\sqrt{1-e^2} \sin \theta}{1-e^2 \cos^2 \theta}. \end{aligned}$$

Question 16 (b) (vi)

Criteria	Marks
• Correct solution	1

Sample answer:

$$\begin{aligned}
 \frac{mrw^2}{mg} &= \frac{\cancel{2T} \sqrt{1-e^2} \sin \theta}{\cancel{(1-e^2 \cos^2 \theta)} \cancel{2T} (1-e^2) \cos \theta} \\
 &= \frac{\sqrt{1-e^2} \sin \theta}{(1-e^2) \cos \theta} \\
 \frac{rw^2}{g} &= \frac{1}{\sqrt{1-e^2}} \tan \theta \\
 \tan \theta &= \frac{rw^2}{g} \sqrt{1-e^2}.
 \end{aligned}$$

Mathematics Extension 2

2013 HSC Examination Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	8.0	H5, E8
2	1	3.2	E4
3	1	2.2	E3
4	1	7.5	E4
5	1	2.5	E3
6	1	4.1	E8
7	1	6.3	E5
8	1	5.1	E7
9	1	1.5	E6
10	1	8.0	HE3, E2

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	1	2.1	E3
11 (a) (ii)	2	2.2	E3
11 (a) (iii)	2	2.4	E3
11 (b)	2	7.6	E4
11 (c)	2	2.1	E3
11 (d)	3	4.1	E8
11 (e)	3	2.5	E3
12 (a)	4	4.1	E8
12 (b)	2	1.8	E4, E6
12 (c)	4	4.1, 5.1	E7, E8
12 (d) (i)	2	3.3	E4
12 (d) (ii)	2	3.3, 8.1	PE3, E3, E4
12 (d) (iii)	1	3.3	E3, E4
13 (a) (i)	3	4.1	E8
13 (a) (ii)	2	4.1	E8
13 (b) (i)	2	1.7	E6
13 (b) (ii)	3	1.8	E6
13 (c) (i)	2	8.0, 8.1	E2
13 (c) (ii)	2	8.0, 8.1	E2
13 (c) (iii)	1	8.0, 8.1	E2
14 (a)	3	4.1, 8.3	PE3, E2
14 (b)	3	2.1, 2.2, 8.2	HE2, E2, E3

Question	Marks	Content	Syllabus outcomes
14 (c) (i)	1	8.0	HE3, E2
14 (c) (ii)	2	4.1	E8
14 (d) (i)	1	8.1	P4, H5
14 (d) (ii)	1	8.1	PE3, E2
14 (d) (iii)	2	8.1	H5, E2
14 (d) (iv)	2	8.1	PE3, E2
15 (a)	3	2.2	E3
15 (b) (i)	2	7.2, 8.0	PE3, E4
15 (b) (ii)	1	7.2, 8.0	PE3, E4
15 (c) (i)	1	8.0	HE3, E2
15 (c) (ii)	2	8.0	HE3, E2
15 (d) (i)	1	6.2	E5
15 (d) (ii)	3	6.2	E5
15 (d) (iii)	2	6.2	E5
16 (a) (i)	2	8.0	H6, E6
16 (a) (ii)	1	8.3	PE3, E4
16 (a) (iii)	2	8.3	PE3, E4
16 (b) (i)	1	3.1	E3, E4
16 (b) (ii)	1	3.1	E3, E4
16 (b) (iii)	2	8.0	H5
16 (b) (iv)	2	6.3	E5
16 (b) (v)	3	6.3	E5
16 (b) (vi)	1	6.3	E5